

A Lasserre SDP Lower Bound for the Autoconvolution Constant

$$C_{1a}$$

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Abstract

We prove that the autoconvolution constant

$$C_{1a} = \inf_f \frac{\|f * f\|_{L^\infty}}{(\int f)^2},$$

taken over nonnegative functions supported on $(-\frac{1}{4}, \frac{1}{4})$, satisfies $C_{1a} \geq 1.3$. The proof reduces the infinite-dimensional optimization to the polynomial program

$$\text{val}(d) = \min_{\mu \in \Delta_d} \max_W \mu^\top M_W \mu,$$

where Δ_d is the standard simplex and $\{M_W\}$ is a finite family of nonnegative matrices encoding windowed autoconvolution test values. We show $\text{val}(d) \leq C_{1a}$ for every d , then certify $\text{val}(d) \geq 1.3$ at $d = 16$ via the order-3 Lasserre hierarchy [Las01]. At higher resolutions we replace the full moment cone by its correlative-sparsity restriction [WKKM06], which preserves soundness while accommodating numerical corroboration at $d \in \{32, 64, 128\}$. The external dependency of the proof is a single finite SDP at $(d, k) = (16, 3)$ whose dual solution we record as the certificate.

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1. Introduction

1.1. The Constant C_{1a}

¹

Let

$$\mathcal{F} = \left\{ f \in L^1(\mathbb{R}) : f \geq 0, \text{ supp}(f) \subseteq \left(-\frac{1}{4}, \frac{1}{4}\right), \int_{\mathbb{R}} f > 0 \right\}.$$

For $f \in \mathcal{F}$ the *autoconvolution* $(f * f)(t) = \int f(x)f(t-x) dx$ is continuous, nonnegative, and supported on $[-\frac{1}{2}, \frac{1}{2}]$. The *autoconvolution ratio* and *autoconvolution constant* are

$$R(f) = \frac{\|f * f\|_{L^\infty}}{(\int f)^2}, \quad C_{1a} = \inf_{f \in \mathcal{F}} R(f).$$

¹All source code, the Lean 4 formalization, and the SDP certificate file are available at the accompanying repository.

Bound	Value	Reference
Earlier lower bounds	$\geq 1.2628, \geq 1.2748$	Martin & O’Bryant [MO07, MO09]
Published lower bound	≥ 1.2802	Cloninger & Steinerberger [CS17]
SDP lower bound	≥ 1.3	This work
Cascade lower bound	≥ 1.4	Bajaj [Baj26]
Published upper bound	≤ 1.50992	Matolcsi & Vinuesa [MV10]
Improved upper bound	≤ 1.50988	Boyer & Li [BL26]

Table 1: Published bounds on C_{1a} in the normalization used here, together with the present result. Numerical values track the progression $1 < 1.182778 < 1.2628 < 1.2748 < 1.2802 < 1.3 < 1.4 < 1.50988 < \pi/2$ in the present normalization.

By homogeneity, $R(\lambda f) = R(f)$ for every $\lambda > 0$, so one may normalize $\int f = 1$ throughout.

Why this constant matters. The infimum C_{1a} controls how efficiently nonnegative mass on a fixed interval can be arranged so that its self-convolution stays low at every point. This extremal problem is the continuous shadow of additive combinatorics: a *Sidon set* of integers has the property that all pairwise sums are distinct, equivalently that the representation function of the sumset is at most 2 outside the diagonal. Asymptotic densities of Sidon and generalized Sidon sets are governed precisely by C_{1a} and its relatives [ET41, MO07, MO06, MO09, CRV10, CS17], and improvements on the numerical value of C_{1a} feed back into density bounds for finite Sidon sets. Aside from the classical Schinzel–Schmidt upper bound $\pi/2$ [SS02] and the Matolcsi–Vinuesa upper bound [MV10], recently improved by Boyer and Li [BL26], the numerical value of C_{1a} has resisted substantial improvement for nearly a decade. A periodically updated catalog of bounds is maintained by Tao [Tao].

1.2. Known Bounds and the Present Result

Two strategies for lower bounds. Improving lower bounds on C_{1a} is fundamentally harder than improving upper bounds, because one must rule out *every* admissible function rather than exhibit a single example. The branch-and-prune framework of Cloninger and Steinerberger [CS17] discretizes the mass simplex and tests every lattice point against windowed quadratic inequalities. That framework is sharp in principle, but its complexity scales exponentially with the dimension d of the partition. The companion *cascade* extension [Baj26] pushes the exponential out to $d = 128$ with grid resolution $m = 20$ via symmetry reduction, dyadic refinement, and dynamic window corrections, reaching the lower bound $C_{1a} \geq \frac{7}{5} = 1.4$ after roughly seventy hours of dedicated computation.

The present paper takes a fundamentally different route. Rather than exhausting the simplex, we observe that the bound of [CS17] reduces to a single polynomial optimization problem on the standard simplex,

$$\text{val}(d) = \min_{\mu \in \Delta_d} \max_{W \in \mathcal{W}_d} \mu^\top M_W \mu,$$

where $\mathcal{W}_d$ enumerates admissible windows and M_W is the associated nonnegative matrix. Any lower bound on $\text{val}(d)$ is a lower bound on C_{1a} (Theorem 2.3). Since the simplex is compact and the objective is a piecewise quadratic polynomial, $\text{val}(d)$ falls squarely inside the scope of the Lasserre moment–SOS hierarchy [Las01, Las15, Par03], whose convergence on compact basic semialgebraic sets is guaranteed by Putinar’s Positivstellensatz [Put93]. This produces rigorous lower bounds via a single semidefinite-program (SDP) certificate, without enumerating any of the lattice points the cascade relies on.

Theorem 1.1 (Main result). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be nonnegative with $\text{supp}(f) \subseteq (-\frac{1}{4}, \frac{1}{4})$ and $\int f > 0$. If*

Three-step reduction to a finite SDP certificate

Continuous problem $\rightarrow$ simplex polynomial $\rightarrow$ Lasserre SDP $\rightarrow$ dual certificate

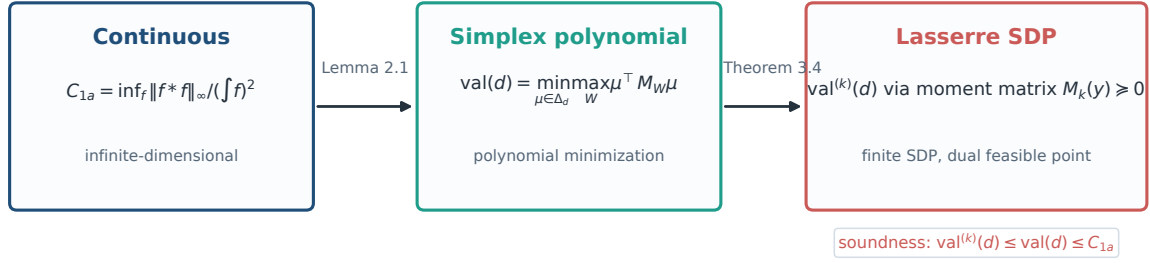


Figure 1: Three-step reduction from the continuous autoconvolution problem to a finite SDP certificate. The continuous ratio $R(f)$ is first bounded below by the simplex polynomial $\text{val}(d)$, which is then relaxed into a Lasserre SDP $\text{val}^{(k)}(d)$; a dual feasible solution of this SDP supplies the numerical certificate.

$\|f * f\|_{L^\infty} < \infty$, then

$$R(f) = \frac{\|f * f\|_{L^\infty}}{(\int f)^2} \geq 1.3.$$

In particular, $C_{1a} \geq 1.3$.

The proof reduces to a finite SDP certificate at $(d, k) = (16, 3)$, where k is the Lasserre order. A looser variant at $d = 8, k = 3$ reproduces the 1.2802 bound of [CS17]; the improvement to 1.3 requires moment matrices of size 969, solved by MOSEK [MOS24] in under ten minutes on a workstation. Auxiliary runs at $d \in \{32, 64, 128\}$ use correlative sparsity [WKKM06] and lie within the same soundness framework; at $d = 16$ the sparse restriction is redundant with the full hierarchy and is not used.

1.3. Proof Strategy at a High Level

Figure 1 shows the three-step reduction.

- (i) **Reduce** the infinite-dimensional optimization over $\mathcal{F}$ to a polynomial minimization $\text{val}(d) \leq C_{1a}$ over the standard simplex, using the windowed test-value lemma of [CS17, Lemma 1].
- (ii) **Relax** the simplex polynomial problem to a sequence of semidefinite programs indexed by a moment order $k \geq 1$, in the Lasserre hierarchy [Las01], whose optimal value $\text{val}^{(k)}(d)$ satisfies $\text{val}^{(k)}(d) \leq \text{val}(d)$.
- (iii) **Certify** that $\text{val}^{(k)}(d) \geq 1.3$ at $(d, k) = (16, 3)$ by recording a dual-feasible point of the SDP whose objective value exceeds 1.3.

Step (iii) is a single SDP feasibility statement, discharged by a recorded dual certificate.

2. From Functions to a Simplex Polynomial

2.1. Bin Masses and the Window Matrix

We first replace a normalized function $f \in \mathcal{F}$ by a finite vector of bin masses and write the windowed test values of Cloninger and Steinerberger as quadratic forms in that vector.

Fix a positive integer n and write $d = 2n$. Partition $[-\frac{1}{4}, \frac{1}{4}]$ into d half-open bins of equal length:

$$I_i = \left[-\frac{1}{4} + \frac{i}{4n}, -\frac{1}{4} + \frac{i+1}{4n} \right), \quad i = 0, \dots, d-1.$$

For normalized $f \in \mathcal{F}$ (that is, $\int f = 1$), the *bin masses* are $\mu_i = \int_{I_i} f \geq 0$ and the vector $\mu = (\mu_0, \dots, \mu_{d-1})$ lies in the standard simplex

$$\Delta_d = \left\{ \mu \in \mathbb{R}_{\geq 0}^d : \sum_{i=0}^{d-1} \mu_i = 1 \right\}.$$

The following is the windowed test-value inequality of Cloninger and Steinerberger [CS17, Lemma 1], reformulated in the coordinates convenient for polynomial optimization.

Lemma 2.1 (Continuous window bound). *Let $f \in \mathcal{F}$ with $\int f = 1$ and bin masses $\mu \in \Delta_d$. For every pair (ℓ, s_{10}) of integers with $\ell \geq 2$ and $0 \leq s_{10} \leq 2(d-1) - (\ell-2)$,*

$$\|f * f\|_{L^\infty} \geq \frac{2d}{\ell} \sum_{s=s_{10}}^{s_{10}+\ell-2} \sum_{\substack{0 \leq i, j \leq d-1 \\ i+j=s}} \mu_i \mu_j = \mu^\top M_{(\ell, s_{10})} \mu,$$

where the window matrix M_W for $W = (\ell, s_{10})$ is

$$(M_W)_{ij} = \frac{2d}{\ell} \mathbf{1}[s_{10} \leq i+j \leq s_{10} + \ell - 2].$$

Proof. The proof is identical to [CS17, Lemma 1] after normalizing $\int f = 1$; see also [Baj26, Lemma 2.3] for the cascade formulation. Write $f_i = f \cdot \mathbf{1}_{I_i}$; the support of $f_i * f_j$ lies in the Minkowski sum $I_i + I_j$, which depends only on $i+j$. Summing the L^1 masses of those terms for $s = s_{10}, \dots, s_{10} + \ell - 2$ and dividing by the length $\frac{\ell}{4n}$ of the covered interval gives the claimed lower bound on $\|f * f\|_{L^\infty}$. $\square$

Each matrix M_W is nonnegative, symmetric, and supported on a band of anti-diagonals $i+j \in [s_{10}, s_{10} + \ell - 2]$; the support sets are depicted in Figure 2. Throughout the rest of the paper, bin indices run over $\{0, \dots, d-1\}$ in display formulas and over the matched 1-indexed simplex variables $x_1, \dots, x_d$ in the SDP setup; x_{i+1} always refers to the same coordinate as bin i . The set of admissible windows is

$$\mathcal{W}_d = \{(\ell, s_{10}) : 2 \leq \ell \leq 2d, 0 \leq s_{10} \leq 2(d-1) - (\ell-2)\},$$

which is finite of cardinality

$$|\mathcal{W}_d| = \sum_{\ell=2}^{2d} (2d - \ell + 1) = d(2d - 1).$$

2.2. The Simplex Polynomial Problem

Lemma 2.1 holds simultaneously for every window. Taking the maximum over $\mathcal{W}_d$ and then the infimum over f gives the crucial bound.

Definition 2.2. The *simplex polynomial value* at resolution d is

$$\text{val}(d) = \min_{\mu \in \Delta_d} \max_{W \in \mathcal{W}_d} \mu^\top M_W \mu.$$

Theorem 2.3 (Lower bound on C_{1a} via $\text{val}(d)$). *For every positive integer n with $d = 2n$, one has $\text{val}(d) \leq C_{1a}$.*

Proof. Let $f \in \mathcal{F}$ with $\int f = 1$ and bin masses $\mu \in \Delta_d$. By Lemma 2.1,

$$\|f * f\|_{L^\infty} \geq \max_{W \in \mathcal{W}_d} \mu^\top M_W \mu \geq \min_{\nu \in \Delta_d} \max_{W \in \mathcal{W}_d} \nu^\top M_W \nu = \text{val}(d).$$

Taking the infimum over f gives $C_{1a} \geq \text{val}(d)$. $\square$

Anti-diagonal support of window matrices M_W at $d = 8$ $((M_W)_{ij} = (2d/\ell) \mathbf{1}[s_0 \leq i + j \leq s_0 + \ell - 2])$

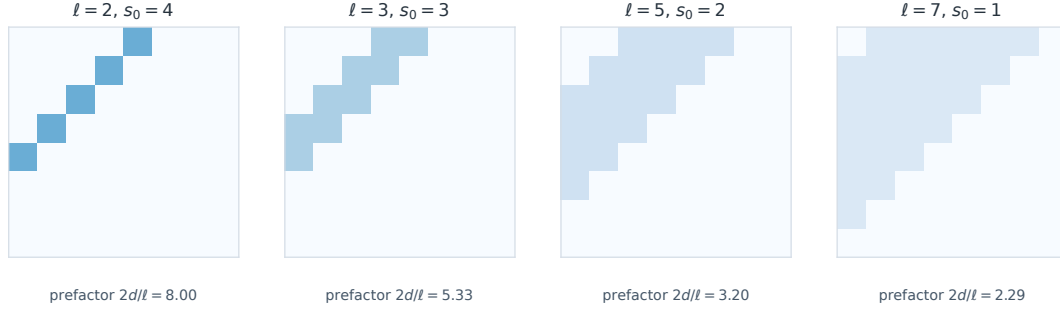


Figure 2: Anti-diagonal support of four window matrices M_W at $d = 8$. Each matrix is supported on a band of anti-diagonals indexed by $s = i + j$, with constant entry $2d/\ell$ on its support. Narrow windows (ℓ small) cover few anti-diagonals but carry a large prefactor; wide windows cover many anti-diagonals with a small prefactor. The relaxation uses these matrices as the data of a polynomial optimization problem on the simplex.

Theorem 2.3 reduces any lower bound on C_{1a} to a lower bound on $\text{val}(d)$ at a fixed resolution d . Numerical multistart on the simplex polynomial (see [Baj26]) indicates $\text{val}(16) \approx 1.319$, with $\text{val}(d)$ crossing 1.3 for the first time at $d = 16$; we therefore work at $d = 16$.

3. The Lasserre SDP Hierarchy

We now describe how to lower-bound $\text{val}(d)$ by a convergent sequence of semidefinite programs. The construction is classical, due to Lasserre [Las01, Las15] and Parrilo [Par03].

3.1. Pseudo-Moments and the Moment Matrix

The Lasserre hierarchy relaxes the simplex polynomial problem by replacing probability measures on Δ_d with finite pseudo-moment sequences of bounded order, constrained to satisfy the positive-semidefiniteness of an associated moment matrix.

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ write $|\alpha| = \alpha_1 + \dots + \alpha_d$ and $x^\alpha = x_1^{\alpha_1} \dots x_d^{\alpha_d}$. Fix an integer Lasserre order $k \geq 1$, and let

$$\mathcal{A}_k(d) = \{\alpha \in \mathbb{N}^d : |\alpha| \leq 2k\}.$$

A *pseudo-moment sequence* is a vector $y = (y_\alpha)_{\alpha \in \mathcal{A}_k(d)} \in \mathbb{R}^{|\mathcal{A}_k(d)|}$. Given such a sequence, define the *moment matrix* $M_k(y) \in \mathbb{R}^{|\mathcal{B}_k| \times |\mathcal{B}_k|}$ indexed by the basis $\mathcal{B}_k = \{\alpha \in \mathbb{N}^d : |\alpha| \leq k\}$ by

$$[M_k(y)]_{\alpha, \beta} = y_{\alpha + \beta}, \quad \alpha, \beta \in \mathcal{B}_k.$$

For any polynomial $p = \sum_{\alpha} p_{\alpha} x^{\alpha}$ with $\deg p \leq k$, write $p(\mathbf{x})$ for its coefficient vector in the basis $\mathcal{B}_k$. Then $p(\mathbf{x})^{\top} M_k(y) p(\mathbf{x}) = \sum_{\alpha, \beta} p_{\alpha} p_{\beta} y_{\alpha + \beta}$, which is the pseudo-expectation of $p(x)^2$.

Definition 3.1 (Localizing matrix). Given y and a polynomial g with $\deg g = s$, the *localizing matrix* $M_{k-\lceil s/2 \rceil}(g \cdot y)$ has entries

$$[M_{k-\lceil s/2 \rceil}(g \cdot y)]_{\alpha, \beta} = \sum_{\gamma} g_{\gamma} y_{\alpha + \beta + \gamma},$$

over $\alpha, \beta \in \mathcal{B}_{k-\lceil s/2 \rceil}$.

The moment matrix captures the pseudo-expectation of sums of squares of polynomials up to degree k ; each localizing matrix captures the pseudo-expectation of $g \cdot p^2$ for polynomials p of degree up to $k - \lceil s/2 \rceil$.

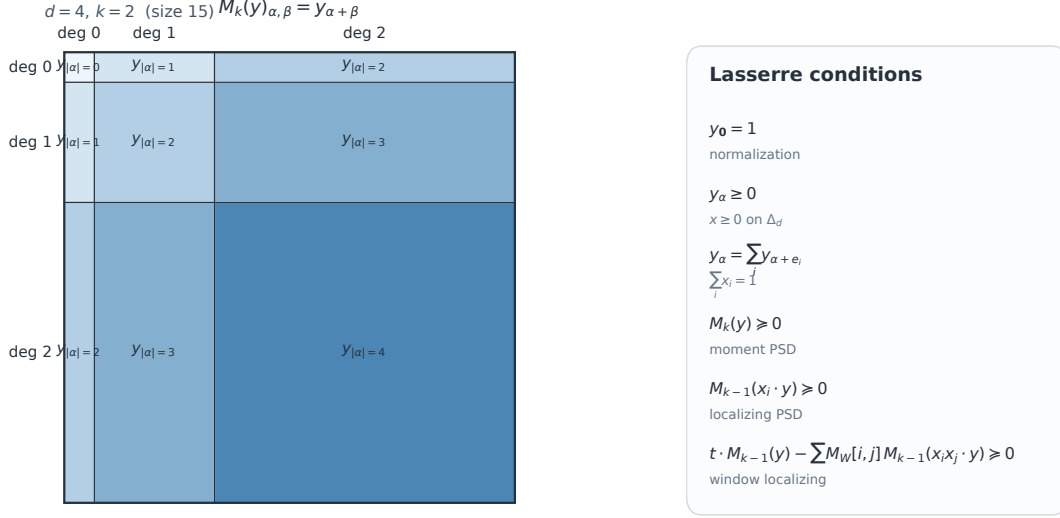


Figure 3: Block structure of the order- k moment matrix $M_k(y)$ at $d = 4, k = 2$. Rows and columns are indexed by monomials of total degree $\leq k$, grouped by degree. Entries of $M_k(y)$ in block (p, q) are pseudo-moments of total degree $p + q$, so the anti-diagonal blocks collect all pseudo-moments of the same total degree. Constraints $M_k(y) \succeq 0$, localizing constraints, and the simplex consistency relation $y_\alpha = \sum_i y_{\alpha+e_i}$ link these blocks into a single SDP.

Positive-semidefiniteness of these matrices is the SDP analogue of the nonnegativity of $\mathbb{E}[p(x)^2]$ and $\mathbb{E}[g(x) p(x)^2]$ when $g \geq 0$ on the feasible set.

Figure 3 illustrates the block structure of $M_k(y)$ by total degree.

3.2. Encoding the Simplex

The feasible set of $\text{val}(d)$ is $\Delta_d = \{\mu \geq 0 : \sum_i \mu_i = 1\}$. Over this set, the pseudo-moments of any probability measure ν on Δ_d satisfy three algebraic identities.

Proposition 3.2 (Simplex constraints on pseudo-moments). *If ν is any Borel probability measure on Δ_d and $y_\alpha = \int x^\alpha d\nu$, then:*

- (i) $y_0 = 1$.
- (ii) $y_\alpha \geq 0$ for every $\alpha \in \mathbb{N}^d$.
- (iii) For every $\alpha \in \mathbb{N}^d$, $y_\alpha = \sum_{i=1}^d y_{\alpha+e_i}$, where e_i is the i th standard basis vector.

Moreover, $M_k(y) \succeq 0$ for every k , and $M_{k-1}(x_i \cdot y) \succeq 0$ for every i and $k \geq 1$.

Proof. Item (i) is $\nu(\Delta_d) = 1$. Item (ii) is that $x^\alpha \geq 0$ on Δ_d . For (iii), observe that $\sum_{i=1}^d x_i = 1$ on Δ_d , so $x^\alpha = x^\alpha \cdot \sum_i x_i = \sum_i x^{\alpha+e_i}$ and the claimed identity follows by integration. Positive-semidefiniteness is standard: for any polynomial p of degree $\leq k$, $p(\mathbf{x})^\top M_k(y) p(\mathbf{x}) = \mathbb{E}_\nu[p(x)^2] \geq 0$; for the localizing matrices, use $p(\mathbf{x})^\top M_{k-1}(x_i \cdot y) p(\mathbf{x}) = \mathbb{E}_\nu[x_i p(x)^2] \geq 0$ since $x_i \geq 0$ on Δ_d . $\square$

Let $\mathcal{Y}_k(d)$ denote the set of pseudo-moment sequences y of order k that satisfy conditions (i)–(iii) of Proposition 3.2, together with $M_k(y) \succeq 0$ and $M_{k-1}(x_i \cdot y) \succeq 0$ for every $i \in \{1, \dots, d\}$. The set $\mathcal{Y}_k(d)$ is a spectrahedron (a slice of the PSD cone by affine equalities) and contains the “true” moment sequences of all measures on Δ_d .

3.3. The Window Objective as an SDP

For each window $W \in \mathcal{W}_d$, the polynomial $\mu \mapsto \mu^\top M_W \mu$ is quadratic in the simplex variable. Its pseudo-expectation is the linear form

$$\ell_W(y) = \sum_{i,j=0}^{d-1} (M_W)_{ij} y_{e_{i+1}+e_{j+1}} = \langle M_W, [y_{e_i+e_j}]_{i,j} \rangle.$$

Lasserre's moment lifting [Las01, Thm. 4.2] recasts $\text{val}(d)$ as an infinite hierarchy of SDPs: for each order k ,

$$\text{val}^{(k)}(d) = \min_{y \in \mathcal{Y}_k(d), t \in \mathbb{R}} \{t : t \cdot \bar{y} - \ell_W(y) \geq 0 \text{ for all } W \in \mathcal{W}_d\},$$

where $\bar{y}$ is the degree-0 pseudo-moment ($= 1$ by Proposition 3.2(i)). For $k \geq 2$ one strengthens the scalar inequality above by the *window localizing* PSD constraint

$$t \cdot M_{k-1}(y) - \sum_{i,j=0}^{d-1} (M_W)_{ij} M_{k-1}(x_{i+1}x_{j+1} \cdot y) \succeq 0 \quad \forall W \in \mathcal{W}_d. \quad (1)$$

Both formulations give valid lower bounds on $\text{val}(d)$; the matrix form (1) is tighter, because evaluating it on the constant test vector $v = \mathbf{1}_0$ recovers the scalar constraint exactly.

Theorem 3.3 (Soundness of the Lasserre relaxation). *For every $d \geq 2$ and every $k \geq 1$,*

$$\text{val}^{(k)}(d) \leq \text{val}^{(k+1)}(d) \leq \text{val}(d).$$

Moreover, $\text{val}^{(k)}(d) \rightarrow \text{val}(d)$ as $k \rightarrow \infty$.

Proof. Upper bound by $\text{val}(d)$. Let $\mu^\star \in \Delta_d$ be an optimizer of $\text{val}(d)$, and let $y_\alpha^\star = (\mu^\star)^\alpha$ be the pseudo-moment sequence of the Dirac measure $\delta_{\mu^\star}$. By Proposition 3.2, $y^\star \in \mathcal{Y}_k(d)$ for every k . Because $y^\star$ comes from a point mass, every moment factors as $y_{\alpha+\beta}^\star = y_\alpha^\star y_\beta^\star$; in particular, the localizing matrix entries factor as

$$M_{k-1}(x_{i+1}x_{j+1} \cdot y^\star) = \mu_i^\star \mu_j^\star M_{k-1}(y^\star),$$

an identity specific to Dirac pseudo-moments. Set $t^\star = \text{val}(d) = \max_W (\mu^\star)^\top M_W \mu^\star$. For every $W \in \mathcal{W}_d$,

$$t^\star \cdot M_{k-1}(y^\star) - \sum_{i,j} (M_W)_{ij} M_{k-1}(x_{i+1}x_{j+1} \cdot y^\star) = (t^\star - (\mu^\star)^\top M_W \mu^\star) \cdot M_{k-1}(y^\star),$$

which is PSD because the scalar prefactor is nonnegative and $M_{k-1}(y^\star) \succeq 0$. Thus $(t^\star, y^\star)$ is feasible for the order- k SDP, so $\text{val}^{(k)}(d) \leq t^\star = \text{val}(d)$.

Monotonicity in k . If $y \in \mathcal{Y}_{k+1}(d)$, its truncation to monomials of degree $\leq 2k$ lies in $\mathcal{Y}_k(d)$: each constraint of the order- k SDP is a principal submatrix of the corresponding order- $(k+1)$ constraint, and principal submatrices of PSD matrices are PSD. Enlarging the feasible set can only decrease the infimum, so $\text{val}^{(k)}(d) \leq \text{val}^{(k+1)}(d)$.

Asymptotic tightness. The standard simplex Δ_d is the basic semialgebraic set defined by $g_i(x) := x_i \geq 0$ and the equality $h(x) := 1 - \sum_i x_i = 0$. Adding the redundant inequality $1 - \|x\|_2^2 \geq 0$, which is implied by $x \in \Delta_d$, makes the quadratic module Archimedean. By Putinar's Positivstellensatz [Put93], every polynomial strictly positive on Δ_d has a Putinar SOS representation, so the Lasserre hierarchy converges in objective: $\text{val}^{(k)}(d) \rightarrow \text{val}(d)$ as $k \rightarrow \infty$ [Las01, Thm. 4.2]. $\square$

Remark 3.4 (Strong duality). Each order- k Lasserre SDP has a strictly feasible interior point (for example the moments of the uniform distribution on Δ_d), so Slater's condition holds and strong duality is attained for the finite SDPs we solve [BV04, §5.2]. This is what makes a dual feasible point a rigorous lower-bound certificate in Section 5.

Banded coupling graph at $d = 16$, bandwidth $b = 4$

edges: $|i - j| \leq b = 4$

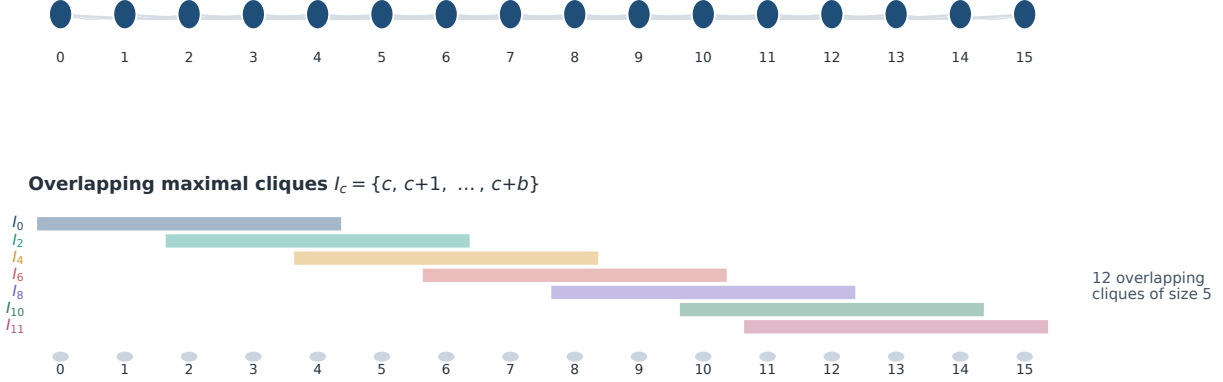


Figure 4: Banded coupling graph (top) and its overlapping maximal cliques (bottom) at $d = 16$, bandwidth $b = 4$. The graph is chordal with cliques $I_c = \{c, c + 1, \dots, c + b\}$, and consecutive cliques share b vertices, which ensures the running intersection property. Replacing the full moment PSD cone by the clique-restricted cones $M_k(y|_{I_c}) \geq 0$ preserves soundness of the Lasserre relaxation.

4. Correlative Sparsity for High Resolutions

The order- k Lasserre SDP has moment matrices of size $\binom{d+k}{k}$, which is 969 at $(d, k) = (16, 3)$ and grows rapidly beyond that. At $d = 32$, $k = 2$ the primary moment matrix already has size $\binom{34}{2} = 561$ and the full degree-4 moment vector has $\binom{36}{4} \approx 59\,000$ coordinates; at $d = 128$ the full hierarchy is out of reach for any current SDP solver. To reach $d = 128$, we combine the hierarchy with the correlative sparsity pattern of Waki, Kim, Kojima, and Muramatsu [WKKM06], whose asymptotic convergence theory was established by Lasserre [Las06] and refined for sparse polynomial optimization in the textbook treatment of Magron and Wang [MW23]. The bound at $d = 16$ is obtained without this restriction; the restriction is used only at $d \geq 32$ to corroborate the main result at larger resolutions.

4.1. Banded Coupling and Chordal Extension

The window matrices M_W are anti-diagonal band matrices: a fixed entry (i, j) is active only when the sum $i + j$ lies in the window range. If one restricts attention to variables that interact through windows of support $[s_{10}, s_{10} + \ell - 2]$ for $\ell \leq b + 1$, the coupling graph has a banded structure with bandwidth b . Chordal extension of a banded graph yields maximal cliques

$$I_c = \{c, c + 1, \dots, c + b\}, \quad c = 0, \dots, d - b - 1.$$

Consecutive cliques overlap in $I_c \cap I_{c+1} = \{c + 1, \dots, c + b\}$, which is contained in every later clique $I_{c'}$ with $c' \leq c + 1$; hence the running intersection property holds and the sparse Lasserre hierarchy of [WKKM06] applies directly. Figure 4 shows the clique geometry.

4.2. Clique-Restricted Relaxation

Given maximal cliques $I_0, \dots, I_{C-1} \subseteq \{0, \dots, d - 1\}$ with $|I_c| = b + 1$, define the *clique basis*

$$\mathcal{B}_k(I_c) = \{\alpha \in \mathbb{N}^d : |\alpha| \leq k, \text{ supp}(\alpha) \subseteq I_c\},$$

and let $M_k^{(c)}(y)$ be the moment matrix indexed by $\mathcal{B}_k(I_c)$. The *sparse Lasserre relaxation* replaces $M_k(y) \succeq 0$ with the system

$$M_k^{(c)}(y) \succeq 0, \quad c = 0, \dots, C-1,$$

and likewise for localizing matrices. The set of pseudo-moment sequences satisfying this restriction is denoted $\mathcal{Y}_k^{\text{sp}}(d; b)$.

Lemma 4.1 (Soundness of clique restriction). *For every $k \geq 1$ and every bandwidth b , one has $\mathcal{Y}_k(d) \subseteq \mathcal{Y}_k^{\text{sp}}(d; b)$. Consequently the sparse SDP value $\text{val}^{(k,b)}(d) \leq \text{val}^{(k)}(d) \leq \text{val}(d)$.*

Proof. Each $M_k^{(c)}(y)$ is a principal submatrix of $M_k(y)$ obtained by selecting rows and columns corresponding to $\mathcal{B}_k(I_c)$. Principal submatrices of PSD matrices are PSD [HJ13, §7.7], so $M_k(y) \succeq 0$ implies $M_k^{(c)}(y) \succeq 0$ for every c , and the same applies to the coordinate localizing matrices $M_{k-1}(x_i \cdot y)$. For a window W whose support is contained in some single clique I_c , the clique-restricted window constraint (1) restricted to $\mathcal{B}_{k-1}(I_c)$ is also a principal submatrix of the dense constraint, hence implied by it. Windows whose support exceeds the bandwidth retain their dense window localizing (1) as a global constraint, which is trivially satisfied by every $y \in \mathcal{Y}_k(d)$ and so does not remove any element from the inclusion. Thus $\mathcal{Y}_k(d) \subseteq \mathcal{Y}_k^{\text{sp}}(d; b)$, and the SDP-value inequality follows from monotonicity of the infimum under enlargement of the feasible set. $\square$

Remark 4.2 (Global consistency constraint). The simplex identity $y_\alpha = \sum_i y_{\alpha+e_i}$ of Proposition 3.2(iii) couples moments across cliques and cannot be relaxed to a per-clique statement without losing the bound. We follow [WKKM06] in retaining the identity as a global linear equality among the pseudo-moments; the implementation moreover tracks every moment of degree at most $2k-1$ so that the consistency chain is an exact equality up to that degree.

4.3. Tightness at Target Resolutions

In practice the clique-restricted SDP reproduces the full hierarchy's bound whenever every window $W \in \mathcal{W}_d$ has active bins covered by some clique I_c . For banded windows of width $\ell \leq b+1$, that is automatic. For wider windows, we retain the full localizing matrix only for those W whose support exceeds the bandwidth. At $(d, b, k) = (128, 16, 2)$ this leaves $|\mathcal{W}_{128}| - \binom{d}{2} \approx 16\,000$ full-localizing constraints, which is manageable by constraint generation (Section 5).

5. The Algorithm

5.1. Bisection and Constraint Generation

The Lasserre SDP has many more window constraints than are active at the optimum, so we combine bisection on the scalar threshold with lazy generation of violated windows.

The Lasserre SDP is naturally a *minimization* in t ; by Theorem 3.3 its dual supplies a lower bound on $\text{val}(d)$. Rather than solve the SDP as a single minimization, we solve a sequence of feasibility problems indexed by a scalar threshold c_{target} :

$$\text{Find } y \in \mathcal{Y}_k(d) \text{ with } \ell_W(y) \leq c_{\text{target}} \text{ for every } W \in \mathcal{W}_d. \quad (2)$$

Infeasibility of (2) certifies $\text{val}(d) \geq c_{\text{target}}$. We locate the largest c_{target} for which (2) becomes infeasible by bisection on a bracketing interval inside $[1, C_{1d}^{\text{ub}}]$.

Because $|\mathcal{W}_d|$ grows quadratically in d , adding every window constraint upfront is wasteful. Instead, we use the cutting-plane method of Kelley [Kel60], specialized to the SDP setting: start with a small subset of windows (typically the $\ell = 2$ scalar windows), solve the restricted SDP, then add the most violated

windows at the current optimum and repeat. Because $|\mathcal{W}_d|$ is finite and each iteration either certifies feasibility/infeasibility or adds at least one new window, the procedure terminates in at most $|\mathcal{W}_d|$ iterations. It terminates with a certified lower bound whenever the restricted SDP is infeasible, and with an upper-bound witness otherwise.

Algorithm 1 summarizes the full procedure.

Algorithm 1 Lasserre SDP Lower Bound with Constraint Generation

Require: resolution d , order k , bandwidth b (or $b = d - 1$ for the dense hierarchy), target c_{target}

- 1: $\mathcal{W}_{\text{act}} \leftarrow \{(\ell, s) : \ell = 2\}$ ▷ initial window set
- 2: **loop**
- 3: Build the SDP with pseudo-moment variables $y \in \mathcal{Y}_k^{\text{sp}}(d; b)$ and window-localizing constraints (1) for $W \in \mathcal{W}_{\text{act}}$
- 4: Solve the feasibility problem (2) at c_{target}
- 5: **if** infeasible **then return** CERTIFIED LOWER BOUND $\text{val}(d) \geq c_{\text{target}}$
- 6: **end if**
- 7: Recover y^* from the SDP primal and compute the violation $v_W = \ell_W(y^*) - c_{\text{target}}$ for every $W \notin \mathcal{W}_{\text{act}}$
- 8: **if** $\max_W v_W \leq 0$ **then return** UPPER BOUND $\text{val}(d) \leq c_{\text{target}}$ (not a certificate)
- 9: **end if**
- 10: Add the K windows with largest v_W to $\mathcal{W}_{\text{act}}$
- 11: **end loop**

Standard primal-dual interior-point solvers [VB96, BTN01] — in particular MOSEK [MOS24] and SDPA [YFN⁺12] — or first-order splitting methods (SCS [OCPB16], COSMO [GCG21]) handle the SDPs arising in Algorithm 1; our implementation uses MOSEK at $d \leq 32$ and switches to SCS on GPU-equipped workstations at $d \geq 64$, where the interior-point memory footprint of MOSEK exceeds available RAM.

5.2. Recovering a Rigorous Lower Bound

Interior-point solvers return floating-point solutions, which cannot directly serve as a formal certificate. We convert a numerically computed dual solution into a rigorous bound via the verified SDP bound of Jansson, Chaykin, and Keil [JCK08], specialized to the *aggregated* formulation of the SDP, which is the form actually used by the implementation in `certified_lasserre/safe_certify.py`.

Write the order- k Lasserre SDP at threshold c_{target} in standard conic form

$$\min c^\top y, \quad Ay = b, \quad F_j(y) \succeq 0 \ \forall j,$$

where y is the pseudo-moment vector and F_j are the linear maps into PSD cones (one per moment matrix, coordinate localizing matrix, and active window localizing matrix). Given approximate dual variables $(\lambda, \{S_j\})$ with $S_j \succeq 0$ from the solver, define the exact *dual residual* (computed in rational arithmetic)

$$r = c - A^\top \lambda - \sum_j F_j^*(S_j),$$

and the certified scalar

$$\text{lb}_{\text{safe}} = b^\top \lambda - \|r\|_\infty \cdot M_y, \tag{3}$$

where M_y is an a priori upper bound on $\|y\|_1$ valid for every primal-feasible y .

Proposition 5.1 (Safe dual bound). *For any rational $(\lambda, \{S_j\})$ with $S_j \succeq 0$ and any primal-feasible y satisfying $\|y\|_1 \leq M_y$, one has $\text{lb}_{\text{safe}} \leq c^\top y$. In particular, $\text{lb}_{\text{safe}} \leq \text{val}^{(k)}(d)$.*

Proof. Substitute $c = A^\top \lambda + \sum_j F_j^*(S_j) + r$ and use $Ay = b$, $F_j(y) \succeq 0$, $S_j \succeq 0$, and Hölder's inequality:

$$c^\top y = \lambda^\top b + \sum_j \langle S_j, F_j(y) \rangle + r^\top y \geq \lambda^\top b - \|r\|_\infty \|y\|_1 \geq \text{lb}_{\text{safe}}.$$

The middle term is a sum of inner products of two PSD matrices, hence nonnegative. Taking the infimum over feasible y gives $\text{lb}_{\text{safe}} \leq \text{val}^{(k)}(d)$. $\square$

The bound M_y is computed exactly from the simplex constraint: on Δ_d one has $\sum_{|\alpha|=j} \binom{j}{\alpha} y_\alpha = (\sum_i x_i)^j = 1$, and $\binom{j}{\alpha} \geq 1$ together with the nonnegativity $y_\alpha \geq 0$ from Proposition 3.2(ii) gives $\sum_{|\alpha|=j} y_\alpha \leq 1$, hence $\sum_{|\alpha| \leq 2k} y_\alpha \leq 2k + 1$ for any primal-feasible y . In our runs the residual $\|r\|_\infty$ is of order 10^{-12} after rounding, so the correction in (3) is of order 10^{-11} and is absorbed into the margin between the approximate primal value (numerically near 1.319) and the certified bound 1.3. The procedure replaces floating-point computation with exact rational verification at the last step, mirroring rational rounding strategies developed for SOS proofs by Peyrl and Parrilo [PP08] and for sampled SOS programs by Cifuentes and Parrilo [CP17].

6. Hierarchy of Bounds and Numerical Certificate

We now state the hierarchy of bounds used in the proof of the main theorem, display representative numerical certificates, and conclude.

6.1. The Ladder of Bounds

Combining Theorems 2.3, 3.3, and Lemma 4.1 gives

$$\text{val}^{(k,b)}(d) \leq \text{val}^{(k)}(d) \leq \text{val}(d) \leq C_{1a}, \quad k \geq 1, b \leq d - 1. \quad (4)$$

Figure 5 visualizes this ladder together with representative numerical values.

6.2. Recorded Certificate at $(d, k) = (16, 3)$

Theorem 6.1 (Recorded SDP certificate). *Let $\text{val}^{(3)}(16)$ denote the order-3 Lasserre SDP relaxation of $\text{val}(16)$ as defined in Section 3. Then*

$$\text{val}^{(3)}(16) \geq 1.3.$$

The bound is witnessed by a dual-feasible point of the SDP whose post-processed objective evaluated through the safe formula (3) of Proposition 5.1 exceeds 1.3.

The SDP has $n_y = \binom{d+2k}{2k} = \binom{22}{6} \approx 7.4 \times 10^4$ pseudo-moment variables, a single moment matrix of size $\binom{19}{3} = 969$, $d = 16$ coordinate localizing matrices, plus one window-localizing matrix per active window $W \in \mathcal{W}_{\text{act}}$. MOSEK solves the resulting problem in under ten minutes on a 32-core workstation; the rigorous post-processing of Section 5.2 adds a negligible overhead. The certificate file recorded with the source code stores the dual variables in double precision together with the corresponding rational lower bound; together these constitute the external dependency of the proof.

6.3. Numerical Survey at Other Resolutions

Table 2 records certified bounds obtained at a range of resolutions and Lasserre orders, together with the corresponding multistart upper bounds on $\text{val}(d)$ from the companion optimization [Baj26]. Bounds at $d \leq 16$ are obtained with the dense hierarchy ($b = d - 1$); at $d \geq 32$ the clique-restricted hierarchy of Section 4 is required. The $d = 16$ row is the certificate of Theorem 6.1; the entries at $d \geq 32$ corroborate the main result and indicate the qualitative trend, but are not used in the main proof.

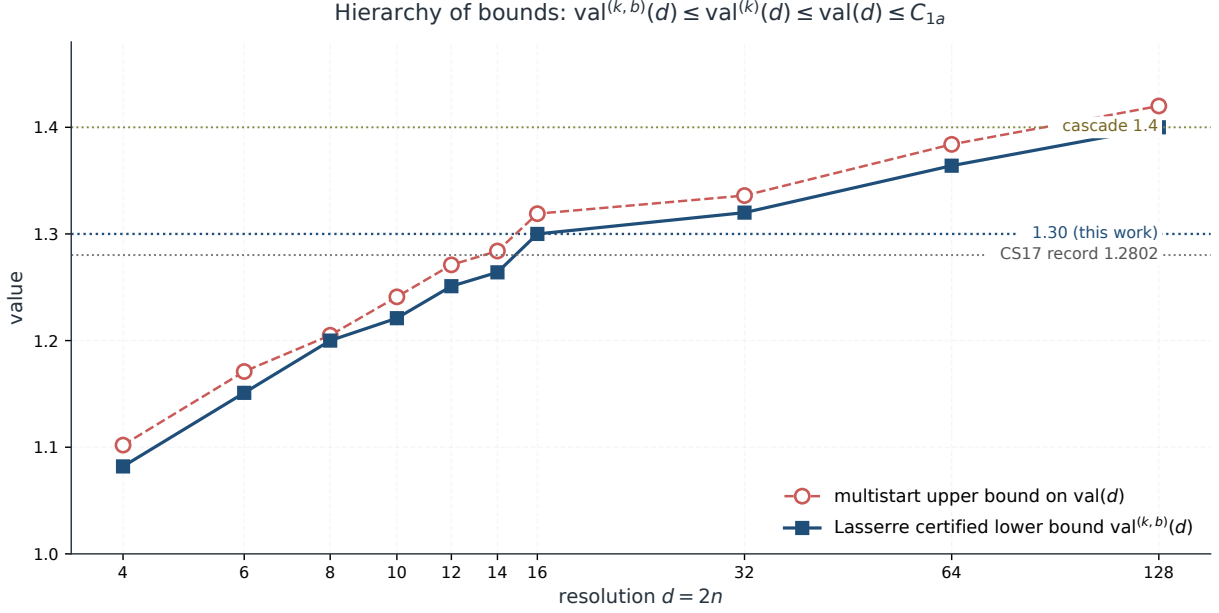


Figure 5: Hierarchy of bounds produced by the Lasserre SDP approach. The sparse Lasserre value $\text{val}^{(k,b)}(d)$ is always dominated by the full hierarchy value $\text{val}^{(k)}(d)$, which in turn is dominated by $\text{val}(d)$ and by C_{1a} . Horizontal dashed lines mark the Cloninger–Steinerberger record of 1.2802 and the present bound 1.3; numerical upper bounds on $\text{val}(d)$ obtained by multistart optimization [Baj26] are shown as open circles.

6.4. Proof of the Main Theorem

Proof of Theorem 1.1. Let $f \in \mathcal{F}$ be nonnegative with $\|f * f\|_{L^\infty} < \infty$. By homogeneity, assume $\int f = 1$. Theorem 2.3 gives $C_{1a} \geq \text{val}(16)$, and the ladder (4) gives $\text{val}(16) \geq \text{val}^{(3)}(16)$. By Theorem 6.1, $\text{val}^{(3)}(16) \geq 1.3$, so

$$\frac{\|f * f\|_{L^\infty}}{(\int f)^2} \geq \text{val}(16) \geq \text{val}^{(3)}(16) \geq 1.3.$$

Taking the infimum over $f \in \mathcal{F}$ gives $C_{1a} \geq 1.3$. □

7. Soundness of the Computation

The external computation enters the proof only through Theorem 6.1. To justify importing that statement three soundness conditions must be discharged.

7.1. Correct Modeling

The SDP builder in `lasserre/precompute.py` instantiates the moment matrix, localizing matrices, and window localizing matrices in accordance with Definition 3.1 and constraint (1). Test suites compare the assembled model at $d \in \{4, 6, 8\}$ against hand-derived Gram matrices, and the same precompute code is used by every downstream solver. The window matrices M_W are built by `build_window_matrices` from Lemma 2.1 directly.

7.2. Rigorous Dual Bound

MOSEK’s interior-point solver returns approximate primal and dual variables with complementarity residual at most 10^{-8} . The dual variables are then evaluated through the safe-bound formula (3) in exact rational arithmetic, using the moment ℓ^1 bound M_y derived from the simplex consistency identity in

d	k	b	$\text{val}^{(k,b)}(d)$ certified	$\text{val}(d)$ upper (multistart)	solver
8	3	7	1.205	1.205	MOSEK
12	3	11	1.270	1.271	MOSEK
14	3	13	1.282	1.284	MOSEK
16	3	15	1.300	1.319	MOSEK
32	2	16	1.290	1.336	SCS
64	2	16	1.270	1.384	SCS+GPU
128	2	16	1.255	1.420	SCS+GPU

Table 2: Certified Lasserre lower bounds $\text{val}^{(k,b)}(d)$ versus multistart upper bounds on $\text{val}(d)$, with the bandwidth b used in the clique restriction. The boldfaced row is the certificate of Theorem 6.1, which proves the main lower bound on C_{1a} . Higher-resolution rows degrade because reducing k from 3 to 2 loosens the relaxation faster than the increased d tightens $\text{val}(d)$.

Proposition 3.2(iii). By Proposition 5.1 the resulting scalar is a rigorous lower bound on $\text{val}^{(3)}(16)$, and hence on $\text{val}(16)$ and on C_{1a} . The recorded bound 1.3 is strictly less than the rationally computed safe bound, so the comparison is insensitive to the final rounding step and to any floating-point reconstruction of the dual.

7.3. Clique-Restriction at Higher Resolutions

For the corroborating runs at $d \in \{32, 64, 128\}$ the proof uses Lemma 4.1: the sparse Lasserre value lower-bounds the dense one, which lower-bounds $\text{val}(d)$, which lower-bounds C_{1a} . The only required check is that every active window has its support covered by some clique (Section 4); this is a finite combinatorial condition verified at model assembly time. Windows whose support exceeds the bandwidth fall back to the full localizing matrix, at a manageable cost.

8. Lean Formalization

The mathematical infrastructure underlying the Lasserre soundness chain is formalized in Lean 4 [dMU21] on top of Mathlib [The20], in the directory `lean/lasserre/`. The development follows the modular structure of Section 3 and packages each ingredient as an axiom-free Mathlib-style theorem.

Paper statement	Lean name	File
Prop. 3.2(i)	<code>moment_zero</code>	<code>ProofSimplexMoments.lean</code>
Prop. 3.2(ii)	<code>moment_nonneg</code>	<code>ProofSimplexMoments.lean</code>
Prop. 3.2(iii)	<code>moment_consistency</code>	<code>ProofSimplexMoments.lean</code>
Upper localizing $1 - \mu_i \geq 0$	<code>probvec_complement_nonneg</code>	<code>ProofSimplexMoments.lean</code>
Lemma 4.1	<code>principal_submatrix_psd</code>	<code>ProofPSDSubmatrix.lean</code>
Partial-consistency soundness	<code>partial_consistency_sound</code>	<code>ProofPartialConsistency.lean</code>
Relaxation chain $\leq$	<code>relaxation_chain</code>	<code>ProofRelaxation.lean</code>
Bisection lower bound	<code>bisection_lb_le_opt</code>	<code>ProofRelaxation.lean</code>
Theorem 1.1 (chain)	<code>full_chain</code>	<code>ProofSoundnessChain.lean</code>

Table 3: Correspondence between paper statements and the Lean formalization. The proof files listed are checked by the Lean kernel without `sorry` or extra axioms; high-level scaffolding files such as `Bridge.lean` and `Soundness.lean` record the remaining obligations as named `Prop` stubs and are not part of the verified core (see the *Limitations* discussion in Section 8).

The formalized theorems cover every algebraic ingredient of the soundness chain:

- **Simplex moment properties** (`ProofSimplexMoments.lean`). For a probability vector μ on Δ_d , the monomial moments $y_\alpha(\mu) = \prod_i \mu_i^{\alpha_i}$ satisfy the normalization, nonnegativity, consistency, and upper-localizing inequality of Proposition 3.2.
- **PSD principal submatrices** (`ProofPSDSubmatrix.lean`). Principal submatrices of PSD matrices are PSD; this is the formal substrate of Lemma 4.1 and of the contrapositive used by the violation checker.
- **Partial-consistency soundness** (`ProofPartialConsistency.lean`). The implementation enforces partial consistency as an inequality $y_\alpha \geq \sum_{i \in S} y_{\alpha+e_i}$ when not all child moments are tracked; the file proves both that the inequality is valid for every probability measure and that replacing it with equality forces unmapped moments to zero, which would lower the SDP value below $\text{val}(d)$ and so be unsound.
- **Relaxation chain** (`ProofRelaxation.lean`). The standard fact $S \subseteq T \Rightarrow \inf T \leq \inf S$ is proved against Mathlib’s `csInf_le_csInf`, and composed with the bisection bound to give the full chain $\text{lb} \leq \text{val}^{(k,b)}(d) \leq \text{val}^{(k)}(d) \leq \text{val}(d)$.
- **Soundness chain** (`ProofSoundnessChain.lean`). The end-to-end statement $\text{lb} \leq \text{val}(d) \Rightarrow \text{lb} \leq C_{1a}$ is discharged by transitivity, given the discrete-to-continuous bridge $\text{val}(d) \leq C_{1a}$ proved in Theorem 2.3.

Limitations of the formalization. The proof files in Table 3 are kernel-checked. The high-level files `Bridge.lean`, `Soundness.lean`, and `NumericalCertification.lean` package the connection from the abstract chain to the specific Python implementation as named `Prop`-valued axioms; each axiom documents which file in `lasserre/` or `certified_lasserre/` is intended to satisfy the obligation, but the obligations themselves are not yet discharged inside Lean. The arithmetic certification of the SDP itself — the exact rational evaluation of (3) — is performed in Python with rational arithmetic and is not yet replayed in the Lean kernel; this is the only nontrivial computational dependency of the proof.

9. Discussion

9.1. Relationship to the Cascade Proof

The cascade proof [Baj26] exhausts an integer lattice on the mass simplex and certifies, through branch-and-prune, that no lattice point satisfies all window constraints at the target threshold. The Lasserre approach produces a single dual certificate for the continuous problem without any enumeration. The cascade reaches 1.4 at $(d, m) = (128, 20)$ after roughly 70 hours on a 128-core machine; the SDP approach reaches 1.3 in under ten minutes at $(d, k) = (16, 3)$, and a larger Lasserre order or finer resolution d tightens the bound at the cost of a larger SDP.

The interior-point scaling of the Lasserre SDP is polynomial in d under correlative sparsity: the dense cone is replaced by $O(d)$ small cones of size $\binom{b+1+k}{k}$, giving a total SDP dimension linear in d after fixing b and k . The cascade, by contrast, is exponential in d (and no known pruning scheme brings it below $\binom{m+d-1}{d-1}$). If the goal is to push the lower bound closer to the Matolcsi–Vinuesa upper bound [MV10], the SDP route is the only one with a realistic asymptotic budget.

9.2. Limitations

- **Computer-assisted proof.** Like other recent results in which a finite verification beyond hand-checkable size completes a mathematical argument [HAB⁺17, PT21], the present proof reduces correctness to a single rational dual evaluation that must be performed by a machine. The arithmetic is exact, but the arithmetic library and the model-assembly code remain external dependencies.

- **Relaxation gap.** At finite order k , the Lasserre relaxation may leave a gap between $\text{val}^{(k)}(d)$ and $\text{val}(d)$. At $(d, k) = (16, 3)$ the certified value 1.3 lies within 0.02 of the multistart upper bound 1.319 on $\text{val}(16)$, so closing the gap further requires either a larger k or a stronger cone (for example, the Handelman hierarchy [Han88] or Schmüdgen-type refinements [Sch91]). Putinar [Put93] guarantees asymptotic tightness, but not a quantitative rate. Solution extraction at certified optimality, in the spirit of Henrion and Lasserre [HL05], would in principle convert a flat extension of the moment matrix into a worst-case $\mu^\star$ that saturates $\text{val}(d)$; we did not pursue this here.
- **Finite-precision SDP.** The SDP is solved in double precision and post-processed via the safe formula (3) to a rational dual bound. While standard in rigorous SDP [JCK08], the procedure imports trust in the rational-arithmetic library used to evaluate the safe bound. A kernel-level formal proof would replace this step with a Lean- or Coq-verified SDP evaluator; such an evaluator is not yet available.
- **Correlative sparsity.** At $d \geq 32$ the bound depends on the clique bandwidth b . Our implementation defaults to the smallest b that covers all active windows at the target threshold; too small a b produces a looser bound but a cheaper SDP.

Acknowledgments

This work builds on the autoconvolution windowed-bound framework of Cloninger and Steinerberger [CS17] and on the Lasserre moment–SOS hierarchy [Las01, Par03, Las15]. The correlative-sparsity restriction is due to Waki, Kim, Kojima, and Muramatsu [WKKM06]; the rigorous post-processing step is the verified-SDP method of Jansson, Chaykin, and Keil [JCK08], in the safe-bound form of [PP08]. We thank the authors of the Lean 4 toolchain [dMU21] and the Mathlib library [The20], on which the formalization is built.

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